

1 Sampling Distribution and OLS Unbiasedness

Usually, we collect multiple distinct datasets (or observation sets) (indexed by j), keeping the independent variable values x_i fixed while observing different outcomes y_{ij} due to random noise. Each dataset yields a unique Ordinary Least Squares (OLS) estimate. Some estimates would be an underestimate of the true line, and some would be an overestimate. However, if we average all these estimates across infinite hypothetical datasets, it converges precisely to the true population parameters β_0 and β_1 , minimizing the effects of localized sampling variance. How reliable a variable is, is determined by the variability in its coefficient- how much does it deviate with respect to the true line (same as the mean)- or in other words, its variance (defined by a term called standard error).

2 Standard Error Formula and Coefficient Derivations

The fundamental building block for the standard error of a sample mean $\hat{\mu}$ across n observations with a constant population variance σ^2 is defined as:

$$\text{Var}(\hat{\mu}) = \text{SE}(\hat{\mu})^2 = \frac{\sigma^2}{n} \quad (1)$$

Let the true simple linear regression model for the j -th observation set at a given x_i value be defined as:

$$y_{ij} = \beta_0 + \beta_1 x_i + \epsilon_{ij} \quad (2)$$

where ϵ_{ij} represents the true unobservable random error term for the j -th dataset at the i -th coordinate. The variance of this structural error term is defined as $\text{Var}(\epsilon_{ij}) = \sigma^2$, which implies that the conditional variance of the response variable is also $\text{Var}(y_{ij} | x_i) = \sigma^2$.

2.1 Derivation of $\text{SE}(\hat{\beta}_1)^2$

The Ordinary Least Squares slope estimator $\hat{\beta}_1$ can be rewritten as a linear combination of the observed outcomes y_{ij} :

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_{ij} - \bar{y}_j)}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x})y_{ij}}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n w_i y_{ij} \quad (3)$$

where the fixed weight for each observation is defined as $w_i = \frac{x_i - \bar{x}}{\sum_{k=1}^n (x_k - \bar{x})^2}$. Assuming that the error terms are completely uncorrelated across observations, we compute the variance of this linear combination as:

$$\text{Var}(\hat{\beta}_1) = \sum_{i=1}^n w_i^2 \text{Var}(y_{ij}) = \sigma^2 \sum_{i=1}^n w_i^2 \quad (4)$$

Evaluating the summation of the squared weights yields:

$$\sum_{i=1}^n w_i^2 = \sum_{i=1}^n \left[\frac{x_i - \bar{x}}{\sum_{k=1}^n (x_k - \bar{x})^2} \right]^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{\left[\sum_{k=1}^n (x_k - \bar{x})^2 \right]^2} = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (5)$$

Substituting this back yields the standard error of the slope coefficient:

$$\text{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (6)$$

2.2 Derivation of $\text{SE}(\hat{\beta}_0)^2$

The intercept estimator is calculated using the sample averages: $\hat{\beta}_0 = \bar{y}_j - \hat{\beta}_1 \bar{x}$. Taking the variance of both sides yields:

$$\text{Var}(\hat{\beta}_0) = \text{Var}(\bar{y}_j) + \bar{x}^2 \text{Var}(\hat{\beta}_1) - 2\bar{x} \text{Cov}(\bar{y}_j, \hat{\beta}_1) \quad (7)$$

The covariance term $\text{Cov}(\bar{y}_j, \hat{\beta}_1)$ evaluates exactly to zero because:

$$\text{Cov} \left(\frac{1}{n} \sum_{i=1}^n y_{ij}, \sum_{k=1}^n w_k y_{kj} \right) = \frac{\sigma^2}{n} \sum_{i=1}^n w_i = \frac{\sigma^2}{n} (0) = 0 \quad (8)$$

Applying the base standard error formula from Equation 1, we know $\text{Var}(\bar{y}_j) = \frac{\sigma^2}{n}$. Combining these elements yields the final expression:

$$\text{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right] \quad (9)$$

3 Estimation in a Single Dataset via RSE

In practical environments, we are given only a single dataset(we are only given the single OLS line which may be biased with respect to the original unknown line)- meaning we cannot directly compute variances across separate observation sets j . This means that we need to find an alternative way to find the variance of the coefficients(or in other words, variance of the error for a particular x). OLS assumes homoscedasticity- the same variance for each x , which means that on an average, our absolute error value for each x can be assumed to tend to a common value(except the lost degrees of freedom). We can thus pool the sample residuals ($e_i = y_i - \hat{y}_i$) across our single sample to construct an unbiased estimate of the unobservable error variance σ^2 using the Residual Standard Error (RSE), where $\text{RSE}^2 = \text{MSE}$.

To mathematically demonstrate the unbiasedness of this approach, we express the sample residuals in terms of the true population errors ϵ_i :

$$e_i = (\epsilon_i - \bar{\epsilon}) - (\hat{\beta}_1 - \beta_1)(x_i - \bar{x}) \quad (10)$$

Squaring and summing these residuals across all n data points gives:

$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (\epsilon_i - \bar{\epsilon})^2 - (\hat{\beta}_1 - \beta_1)^2 \sum_{i=1}^n (x_i - \bar{x})^2 \quad (11)$$

Taking the mathematical expectation (E) across hypothetical samples yields:

$$E \left[\sum_{i=1}^n e_i^2 \right] = E \left[\sum_{i=1}^n (\epsilon_i - \bar{\epsilon})^2 \right] - \sum_{i=1}^n (x_i - \bar{x})^2 \cdot E \left[(\hat{\beta}_1 - \beta_1)^2 \right] \quad (12)$$

Using the properties of sample variance and substituting $\text{Var}(\hat{\beta}_1)$ from Equation 6:

$$E \left[\sum_{i=1}^n e_i^2 \right] = (n-1)\sigma^2 - \sum_{i=1}^n (x_i - \bar{x})^2 \left(\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \quad (13)$$

$$E \left[\sum_{i=1}^n e_i^2 \right] = (n-1)\sigma^2 - 1\sigma^2 = (n-2)\sigma^2 \quad (14)$$

Thus, dividing by the remaining degrees of freedom ($n-2$) provides the unbiased single-dataset estimate for σ^2 :

$$E[\text{MSE}] = E[\text{RSE}^2] = E \left[\frac{\sum_{i=1}^n e_i^2}{n-2} \right] = \sigma^2 \quad (15)$$

Another intuitive approach I thought of was this: We assumed same variance for the estimate, across every x value. We can also observe that while we used our estimate, we have sacrificed on two degrees of freedom (we effectively, have a bias towards two of our n points). Therefore, on an average, we can assume those $n-2$ points to be unbiased towards, and thus averaging on a common magnitude of error- say E . The remaining two terms, having our OLS biased towards them, fit in perfectly. Thus, we can say that:

$$(n-2)\sigma^2 + 2(0)^2 = \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (16)$$

on an average. Thus giving sigma as the above expression 15.

4 Implications for Hypothesis Testing

The magnitude of the standard error acts as the primary gauge for determining the reliability of our regression parameters. The t -statistic used to test the significance of the independent variable ($H_0 : \beta_1 = 0$) is defined as:

$$t = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)} \quad (17)$$

When a dataset contains a high level of unexplainable variation, it generates a larger RSE value. Because RSE acts as the direct proxy for σ in the numerator of our coefficient standard errors, a high RSE forces $SE(\hat{\beta}_1)$ to expand. This inflation directly decreases the absolute magnitude of the t -statistic, dragging it closer to zero. Consequently, this yields a significantly higher p -value, forcing us to conclude that we lack sufficient statistical evidence to state that the parameter is reliably different from zero.